CNN

$(x * w)$

$$\int x(t-a)\, w(a)\, da$$

w

Filter

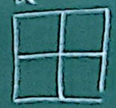
Kernel

fn

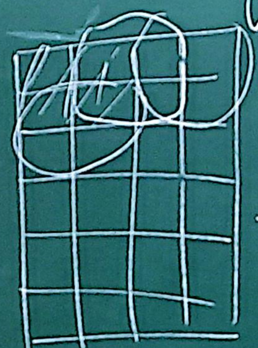
a

$$\int x(a)\, w(t-a)\, da$$

2x2



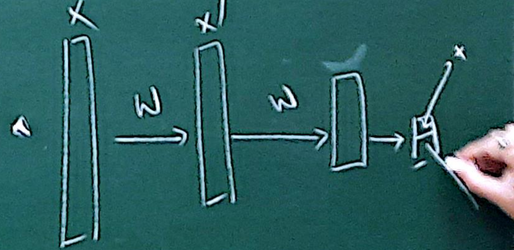
6x4



$g(w^T x + b)$

Pooling

5x3



$W = W_1, W_2, W_3 \ldots\ldots, W_n$

$= E(w_1), E(w_2) \ldots\ldots E(w_n)$

$= Con(\quad)_{1 \times (n \times 5)}$

Word2vec
Glove
FastText

Word Embedding

$E(w_1) = \boxed{\;|\;|\;|\;|\;}$ 5

$E(w_2) = \boxed{\;|\;|\;|\;|\;}$ 5

$U =$

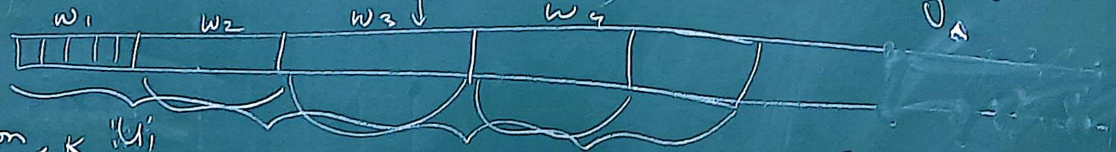
U_1	$1 \times (l \times k)$
U_2	
U_3	

$m \times (l \times k)$

$U|_{l \times k}$ 2

x_i | | | 1

u_1 0 0 0



U = Kernel

l = emb dim

i^{th} position, K U_1, U_2

$$x_i \Big|_{1 \times (l \times k)} = w_i \oplus w_{i+1} \oplus w_{i+2} \oplus \ldots \oplus w_{i+k-1}$$

$\odot$ $\oplus$

$$U\Big|_{1 \times (l \times k)}$$

$$p_i = g(x_i \cdot u)$$

Pooling (p_i)

$$g(x_i \cdot u) = p_i \Big|_{m \times (l \times k)}$$

$W = w_1, w_2, w_3 \ldots\ldots, w_n$

$= E(w_1), E(w_2) \ldots E(w_n)$

Word2vec
Glove
FastText

$= Con($

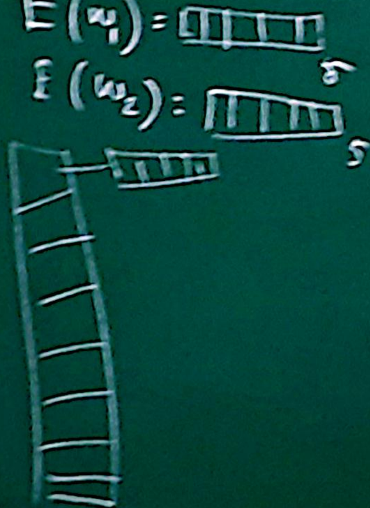
U = Kernel

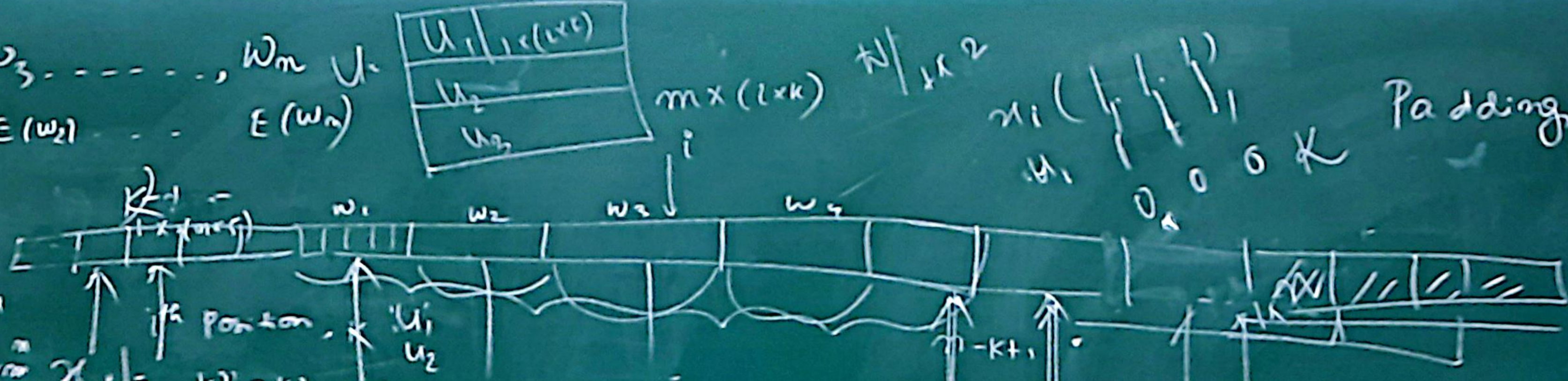
l = emb dim

Word Embedding

$E(w_1) =$

$E(w_2) =$





$m \times (l \times k)$

Padding

$x_i = w_i \oplus w_{i+1} \oplus w_{i+2} \oplus \ldots \oplus w_{i+k-1}$

$1 \times (l \times k)$

$U|_{1 \times (l \times k)}$

$p_i = g(x_i \cdot u)$

Pooling ( )

$g(x_i \cdot u) = p_i$

$m \times (l \times k)$

LIME
SHAPE
3-gram

2-gram
Bi-gram

U ← NER

6×3

1×6
1×3

6×3
max
min
mean

$1 \times (n \times 5)$

w_1 w_2 w_3 w_4

I	w_1					
am	w_2					
going	w_3					
to	w_4					
School	w_5					

$n \times l$

u → I am
u → am going
u → going to
u → to school

w →

5	6	2
0	1	2
2	2	1
1	1	0

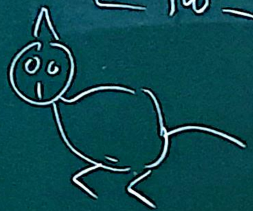
max

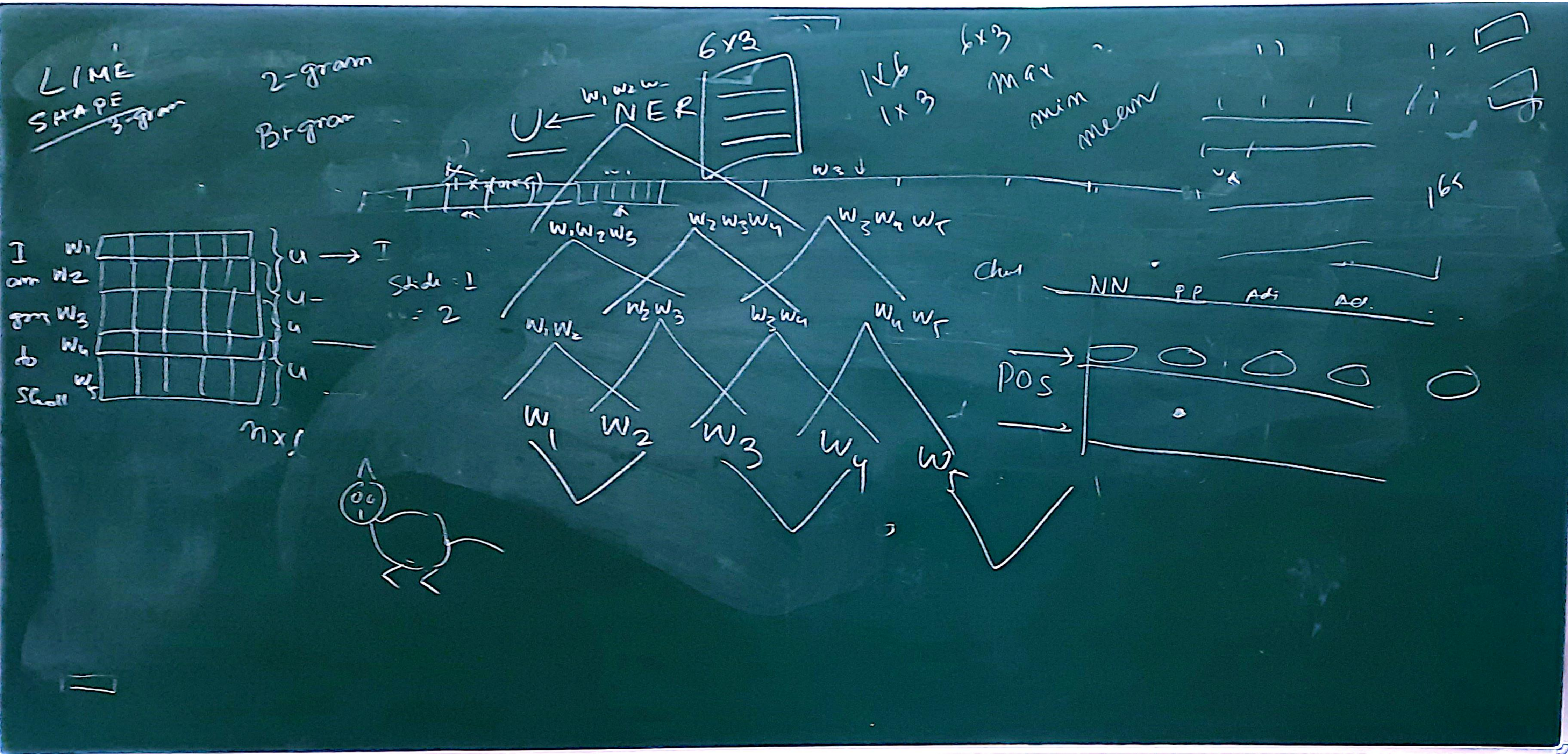
7		

7	6	7
5	2	2

PD
dependency

sub
obj




LIME
SHAPE
3-gram
2-gram
Bigram
6x3
NER
U
1x6
1x3
6x3
max
min
mean
I
am
W1
W2
W3
W4
W5
do
Shall
Stride : 1
: 2
nx1
W1W2W3
W2W3W4
W3W4W5
W1W2
W2W3
W3W4
W4W5
W1
W2
W3
W4
W5
16
Chat
NN
PP
Adj
Adj.
POS

Self
-
Supervised
Learning

Autoregressive model
Next word prediction model

prob distribution over a sequence of tokens

area Chair jumper

a ... 400

$P(am \mid I) = \dfrac{Count(I\ am)}{Count(I\ *)}$

Count(I)

$P(\ a \mid I)$

$P(am \mid I)$

$P(\ldots \mid I)$

I am going ___ ? ___

P(I am go to sch) am

= P(I) P(am) P(... going

P(I) P(am | I) P(go | am) P(to | g ...

W_1 W_2

P(5 7 2 10 15 . .)

50K

may I ___

Unembedding Layer

a

400

Corpus Data

I am going to school now with my mother